

Detecting and Predicting Diabetic Retinopathy

Research Proposal

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1. Problem Statement

Diabetic retinopathy (DR) is a vision-threatening microvascular complication of diabetes that occurs when high blood sugar damages the retinal blood vessels, causing them to swell, leak, or become blocked [1]. In advanced stages, abnormal new blood vessels may also grow on the surface of the retina. Individuals with long-term diabetes or poor blood sugar control are at a higher risk of developing DR, and the likelihood increases with the duration of the disease.

However, diagnosing diabetic retinopathy at an early stage remains a significant challenge because many of the pathological signs, such as microaneurysms, hemorrhages, and vascular abnormalities, can be subtle and difficult to detect in retinal fundus images [2]. Traditional screening methods rely on manual examination by ophthalmologists, which is time-consuming and may not scale effectively for large populations of diabetic patients. Therefore, there is an increasing need for automated systems capable of detecting and predicting diabetic retinopathy from retinal images and associated clinical data, enabling faster screening and supporting clinicians in early diagnosis and disease staging [2].

2. Proposed Algorithms / Models

2.1 Proposed Model 1: CNN-MLP Multimodal Fusion Model

This model integrates retinal fundus images and clinical tabular data for diabetic retinopathy detection. Fundus images are processed using a pretrained convolutional neural network (CNN) such as ResNet50 or EfficientNet to extract visual retinal features. In parallel, patient metadata (e.g., age, diabetes duration, insulin use) are encoded using a Multi-Layer Perceptron (MLP) to generate clinical feature representations. The features from both modalities are fused through feature concatenation and passed to fully connected layers for multi-label disease classification. This architecture provides a simple and effective multimodal framework suitable for datasets like BRSET [3, 4] that contain both images and structured patient data. Multimodal fusion approaches have been shown to improve diagnostic accuracy by combining retinal images with clinical information compared with single-modality models [5].

2.2 Proposed Model 2: Vision Transformer with Clinical Attention Fusion

The second model uses a transformer-based multimodal architecture to capture more complex relationships between retinal images and clinical data. Fundus images are encoded using a Vision Transformer (ViT) or Swin Transformer, which can capture global contextual patterns in retinal structures. The tabular clinical data are processed using a tabular encoder (e.g., TabTransformer or MLP) to produce clinical embeddings. A cross-modal attention layer is then applied to learn interactions between image features and clinical attributes before feeding the fused representation to a multi-label classification head. Transformer-based approaches have demonstrated strong performance in retinal disease detection due to their ability to model long-range dependencies and complex feature relationships in medical images [6].

2.3 Proposed Model 3: Comparative Modeling Framework with XAI

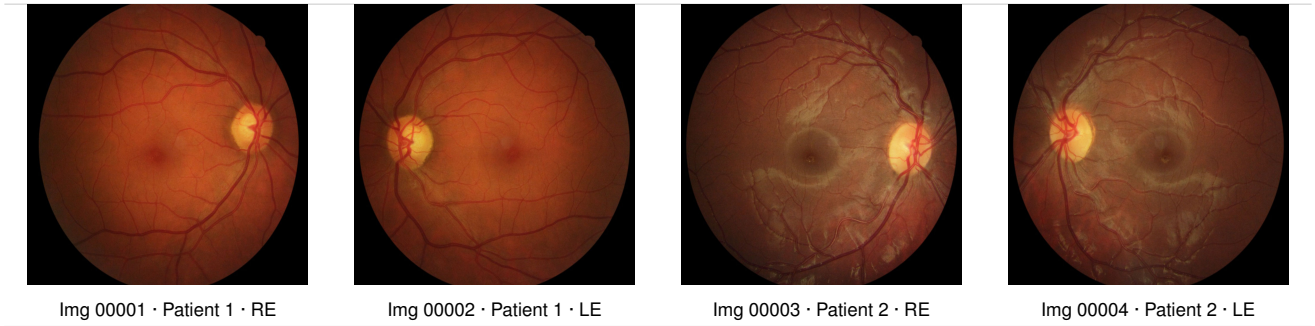
In addition to the multimodal deep learning models, a comparative framework will be implemented using Support Vector Machines (SVM), deep learning models, and convolutional neural networks (CNNs). These models represent a spectrum of approaches, including statistical modeling, ensemble learning, and deep convolutional neural networks. To enhance interpretability, each of the models will be analyzed using four distinct explainable artificial intelligence (XAI) techniques, ensuring that the predictions remain clinically verifiable [8]. The models will be compared using standard evaluation metrics including accuracy, precision, recall, and Area Under the ROC Curve (AUC). This comparative analysis aims to determine which algorithm provides the most transparent and reliable predictive performance for early detection and staging of diabetic

retinopathy [9].

3. Data Description

The dataset comprises high-resolution colour fundus photographs acquired with a Canon CR camera. Each record is labelled by a trained grader across multiple clinical dimensions including diabetic retinopathy severity (DR-SDRG / DR-ICDR), macular status, optic disc morphology, vessel appearance, and overall image quality. The four sample images below illustrate the range of normal and pathological presentations captured in the cohort.

Sample Fundus Images



Annotation Metadata (Snippet)

Image	Patient	Age	Sex	Eye	Diab. Dur.(y)	Insulin	DR-SDRG	DR-ICDR	Macular Edema	Ret. Detach.	Cup-Disc↑	Quality
img00001	1	48	M	R	12	Yes	0	0	0	0	1	Adequate
img00002	1	48	M	L	12	Yes	0	0	0	0	1	Adequate
img00003	2	18	F	R	7	Yes	0	0	0	0	0	Adequate
img00004	2	18	F	L	7	Yes	0	0	0	0	0	Adequate

4	2	18–48	7–12	100%	100%
Sample Images	Unique Patients	Age Range (yrs)	Diabetes Duration (yrs)	Insulin-Dependent	Adequate Quality

4. Related Work

Zeghlache et al. (2025) proposed a longitudinal masked autoencoder (L-MAE) framework for predicting the progression of diabetic retinopathy using self-supervised learning. Their method extends the Transformer-based masked autoencoder architecture by incorporating time-aware positional embeddings and disease-progression-aware masking, allowing the model to capture temporal patterns and pathological changes across follow-up examinations. The study demonstrated that integrating temporal information significantly improves predictive performance compared with conventional deep learning and standard longitudinal Transformer models for diabetic retinopathy progression prediction [6].

Kim et al. (2025) developed machine learning models to predict diabetic retinopathy using clinical data from the Korea National Health and Nutrition Examination Survey. The study compared logistic regression, random forest, decision tree, and XGBoost models. The random forest model showed the highest predictive performance with an AUC of 0.748 [7].

Shahzad et al. (2024) developed a transparent diagnostic framework for diabetic retinopathy using explainable artificial intelligence (XAI) techniques combined with deep learning. Their approach integrates convolutional neural networks with explainability mechanisms to improve interpretability and clinical trust in automated diagnostic systems. The results showed that incorporating explainable AI techniques not only improves diagnostic accuracy but also provides meaningful explanations for model predictions, making the system more suitable for real-world healthcare applications [8].

Bhattacharjee et al. (2021) proposed a machine learning–based method for diabetic retinopathy classification using retinal fundus images. Their approach extracted key retinal features such as exudates, blood vessels, and microaneurysms and applied classifiers including Support Vector Machine, Random Forest, and Naïve Bayes to categorize images into five stages of

diabetic retinopathy. The results showed that the Random Forest classifier achieved the best performance with an accuracy of 76.5%, sensitivity of 77.2%, and specificity of 93.3%, demonstrating the effectiveness of machine learning techniques for automated diabetic retinopathy detection [9].

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